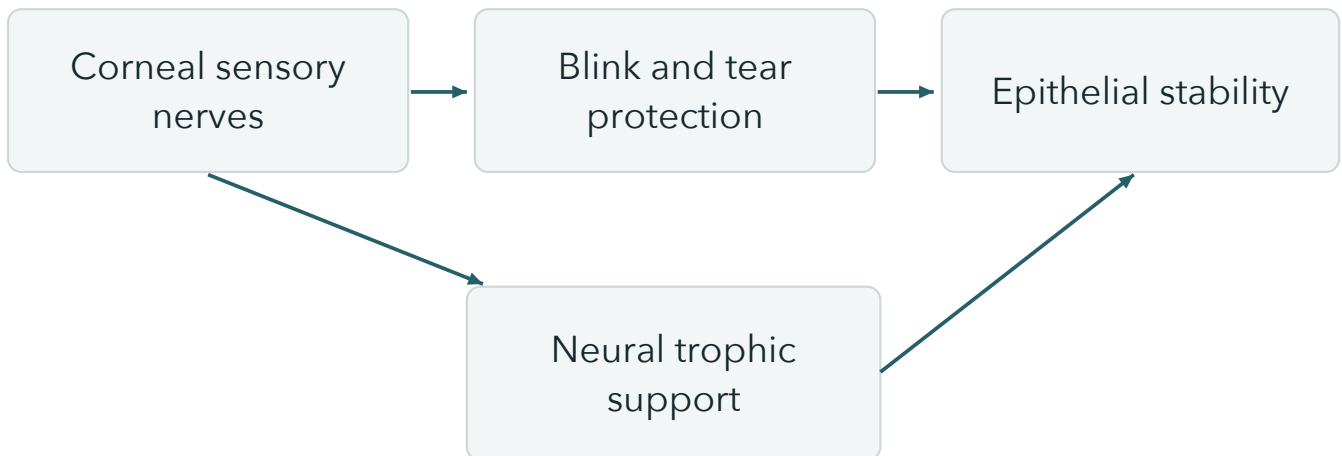


Corneal sensation: neurotrophic failure versus neuropathic pain

Part 2 | A trophic loop can fail quietly

Corneal sensation is part of the examination, not a pain surrogate.



Original schematic of sensory input, protective responses and epithelial support.

Stage pattern	Dominant concern
Surface irregularity	Reduce toxicity and protect epithelium
Persistent epithelial defect	Restore closure and investigate sensation
Melt or perforation	Urgent structural protection

Test sensation before anaesthetic and compare both eyes. Reduced sensation supports neurotrophic disease; assess superimposed infection separately.

Persistent defects have competing explanations

Find the continuing insult as well as the failed repair.

Mechanism	Helpful discriminator
Exposure	Incomplete closure or poor blink
Medication toxicity	Repeated preserved or anaesthetic exposure
Limbal failure	Conjunctivalisation and unstable epithelial phenotype
Reduced trophic support	Impaired sensation with poor healing

Examine lid position and closure, tear coverage, medication use and the limbus. Several mechanisms can coexist. Lubrication may be necessary yet insufficient if the lid remains open or toxic exposure continues. A vascularised irregular epithelium is not proof of active microbial keratitis; sample when infection remains plausible.

Cenegermin evidence answers a healing question

Epithelial closure is not the same endpoint as immediate tectonic stability.

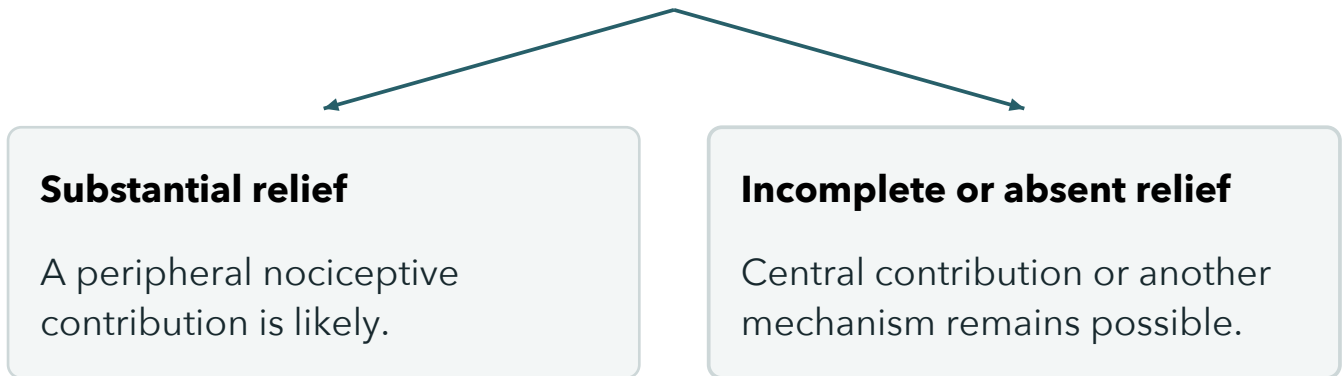
Trial feature	Interpretation
Randomized vehicle comparison	Supports a treatment effect on healing
Persistent defect or ulcer population	Apply to a comparable disease stage
Finite treatment course	Assess closure and later recurrence separately
Structural emergency	Do not wait for a growth-factor course

The pivotal study found more corneal healing with topical recombinant nerve growth factor than vehicle. It does not establish that an imminent perforation can be managed by drug treatment alone, nor that every painful dry eye is neurotrophic. Access, regimen and follow-up require a corneal specialist.

Pain may persist despite a quiet surface

The anaesthetic challenge is a clue, not a definitive localization test.

What happens after supervised topical anaesthetic?



Partial relief can fit mixed peripheral and central sensitisation. Compare pain with surface findings and consider noncorneal causes. Severe symptoms with modest staining do not justify repeated topical anaesthetic at home: toxicity can create a new epithelial problem. A negative challenge alone does not establish a psychiatric cause or exclude ocular pathology.

Neurotization addresses a different timescale

Protect the surface while neural recovery is developing.

- Confirm sensory loss and its cause.
↓
- Protect the epithelium and treat infection or exposure.
↓
- Assess donor nerve and surgical suitability.
↓
- Monitor sensory recovery and surface stability over time.

Corneal neurotization transfers a sensory source to the affected cornea, directly or through a graft. It is a reconstructive strategy with delayed benefit, not an immediate seal for a leaking cornea. Residual exposure, tear dysfunction and inflammation still require treatment, even when sensation improves.